

Original Articles.

FIVE YEARS' ANTI-MALARIA MEASURES ON THE TRAVANCORE TEA COMPANIES' ESTATES.

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WE have been engaged in anti-malaria work over a large area in Travancore for the past six years, and consider that the results we have obtained may be of interest to other people so engaged. By way of introduction may we be permitted to remark that in 1925 one of us, Col. W. W. Clemesha, was consulted by the Travancore Tea Estates Company as to the best way of mitigating the scourge of malaria on several of their estates, some of which had considerable difficulty in showing a profit in unhealthy years. The second writer, Dr. J. H. Moore, is Resident Medical Officer in charge of this group of estates in this part of Travancore. Altogether he has forty-two estates under his care belonging to several different Companies, some of which are malarious and some are not. The senior writer first visited this district in the height of the malaria season of 1925 and has never lost touch with affairs on the estates of the Travancore Tea Estates Company from that time till the present. It is the work on these estates that we particularly wish to discuss. One of us (J. H. M.) has introduced the same methods on many of his large list of estates, and his own figures show that very satisfactory results have been obtained on many of these, varying however with the amount of attention that agents have paid to his recommendations.

Travancore is a large and prosperous Native State situated very near the south end of India. It is located on the western or Malabar shore of India and lies between latitudes 7° to 9° north. In the territory lies a range of hills, which are known as the Western Ghats. These rise to a height of 6,000 ft. or more. There are a large number of upland plateaux among these hills. Tea is grown on these slopes between 2,400 and 4,000 feet altitude; rubber at less. Ravines and mountain valleys are very common on these hills and for the most part they are inclined to be malarious. The flat area of country between the sea and the foot-hills is usually free from the disease.

Climate.

The climate is similar to that in the rest of South India, there being a well-defined dry and

cold weather season, during which time rain is not very heavy but does occur in certain parts when the north-west monsoon prevails. The dry hot weather consists of March, April and May. There is a very strong monsoon period of four months, namely, June, July, August and September, during which about 90 inches of rain usually falls. The further south one goes in the State the less the rainfall is. The monsoon period is essentially cyclonic in nature. The rains are often extremely heavy.

Period of anopheline activity.

In the plateau the period is extremely sharply defined and is spread over four months, namely, the latter half of March, April, May and June, until the rains break. The varieties of anopheles found in the plateau will be discussed later, but as the efficient carriers are nearly all stream breeders they increase in numbers rapidly during the dry and hot months, but are always in danger of being washed away completely when the heavy rains set in. A study of the incidence of malaria in this part of the world shows that reinfection seldom occurs after July. Relapses may, however, go on longer.

The estates about which we wish to make particular remarks are situated in an extremely unhealthy valley, viz., that of the Periyar. The bottom of this depression is 2,500 feet above sea-level. The tops of the hills in the neighbourhood run up to 5,000 feet and even higher. The valley has always had a particularly evil reputation on the score of malaria. Most ravines in this part of the hill country of Travancore are unhealthy, but Periyar valley is particularly so. The reason is that a few miles above the estates is located the large Periyar dam. Most people in India have heard of this. A dam of very considerable height is thrown across the Periyar gorge so that a large lake of 20 square miles is formed with a large catchment area of some 70 square miles. The water of the lake is sent through a tunnel in the hills to the Madura districts for irrigation purposes. As a result of this, for a large number of months in the year the river Periyar below the dam is seldom properly flushed out, and that only when the Periyar lake surpluses over the sill. Consequently the river is nothing else but a chain of pools over a rocky bed with a very slight current, and is a very favourite breeding place for several varieties of anopheles, the particular one which causes most of the fever in the bottom of the valley in this part of the country being *A. culicifacies*. The following other varieties have also been found more or less plentifully:—*A. funestus* var. *listoni*, a few *A. aconitus*, *A. karwari* and the usual wild varieties. Upon the sides of the hills are many ravines, with streams at the bottom of each; some small swampy areas; a large number of

patches of seepage arising from small springs from the hill-sides are also a constant feature. The streams, patches of seepage, etc., produce a large number of *A. maculatus*, *A. listoni* and a few *A. aconitus*, particularly in the grassy edges of streams and swamps. It is obvious, therefore, that with a large number of *A. culicifacies*, *maculatus*, *listoni*—not to mention several of the more doubtful carriers

been fairly well established, though had we had more leisure several others would have claimed attention. As both the writers are paid by commercial companies it has always been our endeavour to adopt the point of view of those who pay us, that is to say to concentrate on measures which are going to reduce financial loss, rather than the more academic problems.

TABLE I.

	1925.			1926.			1927.			1928.			1929.			1930.		
	Percentage free from spleen.	Percentage with enlarged spleen.	Number examined.	Percentage free from spleen.	Percentage with enlarged spleen.	Number examined.	Percentage free from spleen.	Percentage with enlarged spleen.	Number examined.	Percentage free from spleen.	Percentage with enlarged spleen.	Number examined.	Percentage free from spleen.	Percentage with enlarged spleen.	Number examined.	Percentage free from spleen.	Percentage with enlarged spleen.	Number examined.
<i>Munjamullay</i> .—																		
Upper division	72	28	92	86	14	90	74	26	98	91	9	100	92	8	90	100	19	20
Factory	58	42	26	58	42	20	48	52	21	95	5	23	63	32	25	20	79	33
Lower division	78	22	62	84	16	65	92	8	71	97	3	60	96	4	70	97	3	73
<i>Tungamullay</i> .—																		
Dispensary line	29	71	45	90	10	36	80	20	39	Work of dispenser very bad; outbreak.			77	13	48	94	6	50
Augustine line	..	..	..	50	50	10	55	45	12				..	..	31	..	..	26
Karapens	53	47	60	64	36	53	54	46	55	75	25	53	..	98	2	48	..	..
<i>Nellikai</i> .—																		
Top lines	90	10	35	95	5	40	95	5	38	98	2	36	100	..	33	100	..	40
Upper bungalow	78	22	40	85	15	51	100	..	55	89	11	53	93	4	111	94	6	33
Wallardie side	80	20	48	88	12	43	88	12	45	96	4	46	93	2	50	..	..	..
Worst line	18	82	14	82	18	9	..	..	..	..	..	..	..	..	..	..	..	..
<i>Injikadu</i> .—																		
Periyar side of estate.	60	40	55	82	18	50	Measures useless; truly dreadful outbreak; lines abandoned; many deaths.			86	14	104	95	5	90	27	95	5
Bungalow side of estate.	90	10	48	92	8	52				..	..	..	..	..	..	..	..	..
Condemned lines.	..	..	..	75	25	12	82	18	28	..	..	..	..	..	..	..	..	..
<i>Pasumullay</i> .—																		
Whole settlement on hill.	94	6	140	Not examined.			95	5	155	93	7	153	98	2	149	21	93	2
Two lowest lines.	72	28					75	25		..	..	..	..	..	..	..	..	..
Top lines of all	100	..					93	7		..	..	..	..	..	..	..	..	..

Estates that served as a control.
No measures of any kind adopted.

1928.	Chorakolam	..	..	..
	Upper lines	..	50	50
	Lower lines	..	46	54
1929.	Periyar estate	..	60	40
	Periyar bazar	..	75	25

—malaria has every opportunity of spreading if no measures are adopted to combat it. It is these measures rather than the entomological side of the problem which we wish to discuss in this note. Entomology has received a good deal of attention, and all fundamental points have

The table given above speaks for itself: it does not require a great deal of detailed discussion. It is, however, necessary to say that in publishing our results we deal largely with spleen rates. These give a satisfactory basis for estimating the improvement in the health

of a labour force, but they do not tell the whole story. Children who have lived in hyperendemic areas all their lives do not lose enlargement of the spleen readily; we have found from a great deal of experience that it takes about as long to cure an enlargement of the spleen as it does to acquire it. A child of four who has been regularly infected through several months of each year since it was born will have a large spleen, and it will take years of good treatment to reduce it, but it may have little or no fever after measures are started. On the other hand a child living in normal healthy lines, where an outbreak has recently occurred, and who has only had an enlargement of the spleen from four to six weeks, will lose this with decent treatment in a similar period of time. The main point which we wish to accentuate is that, although the rate of recovery as judged by the spleen rate is low, the amount of loss to the Company is recovered much more quickly. It is easy to see how this may happen. With scientific and vigorous anti-malarial measures, serious cases very soon cease to exist. Sixty-six to seventy-five per cent. of the loss occasioned by malaria can be removed in six months. The remaining 33 or 25 per cent. takes much longer time; yet the fall in the spleen rate in twelve months may not be anything very striking. The number of days' work lost to the Company, due to sickness from malaria, bears no relation to the spleen rate; as we have already said the former can easily be divided by about 4 or 5 in six months if measures are scientific and applied with vigour.

As will be seen from the above table we have been fortunate in having two native estates as control for our work, on which no anti-larval measures have ever been carried out. The figures are extremely interesting and represent almost exactly the condition of affairs on our estates when work commenced in 1926.

The senior writer started investigating the problem of these estates in May 1925 at the height of the malaria season. At this time conditions were extremely bad and a few weeks' work enabled one to get a fair idea of the intensity of malaria and the amount of damage done. A large number of coolies used to go to the field to pluck tea, and a fair percentage used to collapse with malaria in the field. The out-turn of work was frequently as low as 25 per cent. and even lower, for weeks together. When it is remembered that during these months there is a great deal of leaf to be plucked, the loss to the Company can easily be imagined. Hundreds of acres of tea were injured through not having sufficient labour to pluck the crop at the proper time. Consequently it is obvious that not only was there a loss in the income of the Company, which in itself is serious enough, but the tea plants were damaged, thereby interfering with their

future productivity; the damage occasioned by this is very difficult to compute. During the period under discussion—May 1925—all the cases were malignant tertian, in the earlier weeks benign tertian certainly occurred. The usual story was as follows:—A coolie sickened with malaria and was very ill from about 8 to 14 days; deaths were not at all uncommon, especially amongst children. When the fever had ceased the patient was completely incapacitated from work for another two weeks, and in most cases convalescence was very slow and lasted about 4 to 8 weeks before the patient was able to do a full day's work. The patient was frequently so reduced in health that he had to be sent home. At the end of May it was no uncommon occurrence to find everybody in a line—men, women and children—with parasites in their blood, having enlargement of the spleen, the whole community having suffered from an attack during the fever season; sometimes three-quarters of the labour force were sick at the same time. The figures in the table cannot give any idea of the actual state of affairs; they are given for divisions or groups of lines, so that it frequently happened that there was one very unhealthy line and 3 or 4 fairly healthy ones further up the hill-side which made the average spleen rate nothing very alarming. An interesting example of what is meant will be seen by the state of Nellikai, where on the Wallardie side of the estate there was 80 per cent. free, 20 per cent. with enlargement, but the nearest line to the stream had only 18 per cent. free, 82 per cent. suffering with enlargement of the spleen.

In this valley cases of malaria commence in the latter end of March; anopheles larvæ in the small breeding places in the more sheltered localities begin to be common in the third or fourth week of February. In the Periyar river itself it is seldom that larvæ became very common until the end of March, or the first or second week of April, but, when the river does start breeding, it pours out thousands of anopheles per mile per night. It was a common occurrence to catch as many as 50 to 70 larvæ in a single ladleful at the height of the season.

Cases usually begin at the end of March and increase in number and virulence till the onset of the monsoon in the early part of June. As the normal rains are heavy in this part of the world, the breeding places are thoroughly scoured out by the burst of monsoon, and conditions begin to improve immediately after the rains set in. In the old days it used to take 4 to 6 weeks for re-infections and serious cases to cease, and, after a bad year, a large number of deaths in July and August from nephritis, due to poisoning of the kidneys with malaria toxin, were reported. In 1927 we had 22 deaths in a few weeks on one estate from this cause.

Such was the picture of the state of affairs when investigation work commenced. By the malaria season of 1926 all arrangements were made for a vigorous campaign of remedial measures, and since that date the picture given above has never been repeated, except in isolated groups of lines in the estates where for one reason or the other the measures failed.

Turning to the other side of the picture, for the last three years we have not had 10 serious cases of malaria like the ones described above; the cases that did occur were mostly Europeans and their servants, and were mild and few and far between. Most of the Europeans were infected at the Club which is located in a very bad situation. Relapses during recent years have almost disappeared. The out-turn of work even in the deadly month of May is now 98 to 99 per cent., and the Company has not lost a pound of tea through the failure of plucking since anti-malarial measures were started. Probably the change in the appearance of the children is the most striking result of the campaign. The death rate per thousand of the population per annum on these five estates is now about 20 or the same as that of an English town.

It is a platitude that one learns most from one's mistakes. During the period of the five years under review we have had two breakdowns, mention of which will be found in the table. The first one was on Injikadu in 1927. There were a series of lines on the top of a spur rather too close, in our opinion, to the Periyar river itself. At my first visit I condemned the 3 lower lines; unfortunately, although it was agreed to pull these down, other accommodation was not available by the next year, consequently coolies were housed in them. Spreading from the river to these lines a very serious outbreak took place in spite of the fact that no other breeding place in the neighbourhood was producing anopheles, and that the children were being carefully treated. The whole of this side became involved in the outbreak; practically everybody suffered from the disease and there were 22 deaths. By the next malaria season the whole of the labour force on this side of the estate were transferred to new lines, constructed on a site selected on the side of a hill, far removed from the Periyar river. Since that date no recurrence of the disease has occurred beyond sporadic cases.

The second instance of break-down occurred on Tungamullay estate. There were not so many fatalities, but it was serious because the health of the labour force suffered severely and took two years to recover. It was due to the inefficient working of the compounder in charge of the treatment and the oiling. He was an elderly man, who, when the writer was present, did very excellent work. Owing to advancing years he appeared to lose his grip on the situation and almost before we realised what

had happened the whole labour force on the estate were involved in an outbreak of the disease, the like of which we had not seen for two years previously. It was an interesting though an expensive occurrence from the Company's point of view. It demonstrated to us the truth of a point which we have always insisted on, viz., that anti-malarial measures must be begun exactly at the right time of the year if there is to be any hope of success. Although the most strenuous efforts were made to cope with the situation after cases began to occur, they only yielded very inadequate results; it was not until the malaria season proper was over that some vestige of health returned to the labour force on this estate, and it was two years before they recovered.

Before discussing the details of the measures employed, we think it necessary to make some preliminary remarks for the benefit of those who may not be very familiar with conditions on tea and rubber estates. The first and most important of these is as follows:—That a community of say 1,000 tea garden coolies, with old people and young children, live under very much better conditions than the ordinary villager. In the first place they get regular pay and hence regular meals. It is very seldom that an ordinary tea garden cooly on a well managed estate shows any sign of suffering from economic stress or under-feeding after he has been three months on the estate. New-comers and their children are frequently half-starved and often are suffering from a considerable degree of hookworm infection on their arrival, and in some cases, though not usually, are badly infected with malaria. Three months' residence on an estate with regular pay, good food and steady work effects a very great improvement in their health. In Travancore and Southern India labour is reasonably well housed, and at any rate it is safe to say that the accommodation provided is much superior to what they are accustomed to in villages.

The second most important point about the tea garden labour force is that they are under discipline. On all well managed estates the word of the superintendent is law. A good superintendent who understands his coolies never asks the impossible, he neither overdrives them nor spoils them; any reasonable orders which he gives are actually carried out with a minimum of difficulty and delay. This constitutes the main difference between a labour force and the civil population. It follows therefore that, when dealing with a labour force, it is possible to carry out a definite campaign against gametocyte carriers; such a thing is practically hopeless with a civil population. Hence anti-malarial measures are much easier to carry out, and much more satisfactory in their results in an organised institution like a tea or rubber estate than they are in villages; because on an estate one can attack

the problem from two different points of view—(a) anti-anopheline and (b) anti-gametocyte—from both flanks, whereas in a civil population it is only possible to do so from one. Any thoughtful person will appreciate the vital importance of these conditions.

We wish now to say a few words about a subject which lies at the root of all anti-malarial measures.

The senior writer of this article has been the recipient of a considerable mathematical education. During a long service in Government employment he arrived at certain conclusions, which he was never able to put to trial; after retirement, when his work lay amongst an organised colony like a tea garden or a rubber estate, opportunities occurred for testing the validity of these early conclusions which were arrived at from a study of the mathematics of the prevention of disease.

A great deal has been written on the mathematics of epidemics generally and malaria in particular by Sir Ronald Ross, Christophers, Gill, and McKendrick on tropical problems and by other well-known writers. The subject is an absorbing one and one which has been apparently entirely overlooked by certain people who have to their credit some very notable success in the field of prevention of disease. All epidemiologists, however, must take cognisance of this aspect of the subject if they wish to get the best out of their efforts. We consider the reason that it has been neglected up to the present is twofold:—(1) That the medical man's training does not predispose him to thinking on mathematical lines; his education is essentially "human" in its nature—that is the reason why it is the finest in existence. Pure mathematics and statistics cannot possibly be called "human"; and (2) that the measures, which are the logical outcome of mathematical argument, are usually entirely inapplicable amongst the civil population; they are not, however, impossible amongst an organised community like a tea garden labour force, a regiment of soldiers, a jail full of convicts, a large boys' or girls' school, etc.

We wish to avoid as far as possible complicated formulæ; we desire to state our position in the simplest possible terms:—

(1) If we could do away with 100 per cent. of the anopheles in an area the spread of the disease would cease, the number of the gametocyte carriers would decline, and the disease would die out.

(2) If we could cure all the gametocyte carriers in any community malaria would likewise disappear, and it would be immaterial how many anopheles there were.

The above are obviously platitudes; in the tropics both are impossible of realisation.

(3) A considerable reduction (66 per cent. to 75 per cent.) in the number of efficient carrier anopheles *only* may have little or no effect on

the number of reinfections if the number of gametocyte carriers in the community is high. The remaining 33 per cent. to 25 per cent. of anopheles readily become infected and are sufficient to disseminate the disease.

(This is well known to all malariologists. Christophers was, we think, the first to point out that half a dozen infected female anopheles are capable of infecting everybody in a line or a cluster of houses.)

(4) Similarly, a considerable reduction (66 per cent. to 75 per cent.) in the number of gametocyte carriers *only* will affect the number of reinfections very little if there are a large number of efficient carrier anopheles present.

(5) Manifestly, therefore, the only really scientific and mathematically correct method of stopping reinfections is to get the maximum reduction in the number of anopheles and the maximum reduction in the number of the gametocyte carriers at the same moment.

If conditions are favourable for an 80 per cent. to 90 per cent. reduction in the number of gametocyte carriers, and 66 per cent. to 75 per cent. reduction in the number of dangerous anopheles, there will be few reinfections; any that do occur are of the nature of accidents; and the progressive increase in the incidence of a disease cannot be maintained by accidents.

(6) It is well known to all students of mathematics that, given certain quantitative conditions, such as sufficient anopheles and sufficient gametocyte carriers per 1,000 of the population, the incidence of a disease must increase; decrease these vital conditions below a certain level (which can be calculated or represented by a line on a graph), and the disease must decline.

We consider that the above simple facts must govern all active measures, and failure to observe these lies at the root of practically all the break-downs in anti-malarial measures throughout the world, including our own.

No one but the veriest tyro ever expects to get a 100 per cent. success in the reduction of the number of anopheles in any locality. In the pre-plasmoquine days the reduction in the number of gametocyte carriers was a very difficult piece of work; long courses of quinine seemed to increase the number of gametocytes available in the blood on some occasions. It also follows from the above that the position taken up by certain malariologists that anti-larval measures are the only successful way of stopping malaria is wrong. It may be admitted that the anopheles being the active or living partner, while the gametocyte carrier is the passive or sleeping partner, it is better to begin with the anopheles; but our point is that it is mathematically wrong to neglect the gametocyte carrier. Further, when only anti-larval measures are used, the results are much too slow for the commercial world, and complete

failures are fairly common. Owing to the activity of a small number of anopheles it may take years to reduce the number of gametocyte carriers below the critical level. The only thing to be said, for those who pin their faith absolutely to anti-larval measures alone, is that anti-gametocyte measures are impossible in certain communities, notably the civil population, and until recent years anti-gametocyte measures were not a success, because we had no drug that would remove gametocytes easily from the blood of an individual. All this is now changed. Consequently, both on the grounds of common sense and pure mathematics, there is only one right way of reducing malaria—that is to reduce both the number of vectors (anopheles) and the number of gametocyte carriers below a certain line simultaneously. This is now a comparatively simple matter, though it was much more difficult in 1925 when this work commenced.

There is one more interesting conclusion from the above argument; that is that there is nothing so foolish as half-hearted, slipshod, anti-malarial measures. These are worse than useless; they may cost a considerable sum of money and produce little or no improvement. It is more logical to do nothing at all than to waste money on half measures. The malarialogist must be a "whole-hogger," or he had better take to some other branch of medicine.

On the subject of anti-larval measures we have had much experience. The results, or rather the degree of success obtained, depend on many factors. These may be briefly enumerated as follows:—

(1) *The nature of the breeding place* influences the possibility of satisfactory control. Small streams, small patches of seepage can be very easily dealt with; large rivers, large swamps, etc., it may be impossible to do anything for on the ground of expense. Remember we are looking at the problem from the point of view of a business concern which must pay dividends, not from that of Government with large resources.

(2) *The size of the area controlled*; manifestly the larger the area efficiently dealt with, the greater the reduction of anopheline population will be in the centre. The area controlled is usually governed by the expense involved. Various authorities have laid down distances which have been found in practice to be reasonably satisfactory: thus, Sir Malcolm Watson says that 40 chains or half a mile is enough for *A. maculatus* breeding places in the Federated Malay States. With *A. culicifacies* as the main carrier we cannot lay down any such distance. We know of many instances where this carrier has certainly flown two miles, and it is suspected of having done much greater distances in a sparsely populated country, when in search of food.

(3) *The rate of invasion from outside*. Invasion from outside invariably takes place. In our own estates where every drop of water was carefully oiled over a large area, we were always able to trap a few *A. culicifacies* in the lines, which had come in from outside. They were not in sufficient numbers in any particular year to cause cases. In some places we have met with it is quite useless to spend money on oiling; we probably only prevented from hatching out some 5 to 10 per cent. of the total anopheline population of the estate, the other ninety per cent. came in from outside. It is only fair to state that these cases are not very numerous and nearly all possess the same general characteristics. There are none in the Periyar valley. In many Assam estates invasion from outside will be found a very serious factor.

(4) *The cyclic variation in the number of the carrier anopheles*. About this we do not propose to say much at present. It will be made the subject of a separate communication when all our researches are complete. Briefly we mean that one species of anopheles, like all other flying insects throughout the world, is subject to variation in numbers from year to year; it follows a definite cycle of approximately 6 to 8 years. The phenomenon has been observed over very large areas such as the whole of Ceylon, and in parts of Travancore in 1929-30. Consequently well-thought-out measures, which have yielded perfectly satisfactory results in the control of anopheles for 5 or 6 years, may entirely break down on the seventh.

From the above it is obvious that anti-anopheline measures, though absolutely necessary for success in most instances, require great care and constant scientific study in any given locality.

Turning now to anti-gametocyte measures, it must be admitted that the discovery of plasmoquine has put an entirely different complexion on this side of the problem. In the adult Tamil cooly it was formerly a practical impossibility to cure him of his gametocytes. The only way it could be done was a prolonged course of either quinine or cinchona febrifuge extending over 2 to 3 calendar months. No cooly will put up with the annoyance of this procedure. Indeed we can go further and say that very few Europeans would persevere sufficiently long with the treatment; they would rather have an occasional attack of malaria than endure the remedy. With little children matters are easier. In the first place it is thoroughly well known that in a community it is usually the small children who are gametocyte carriers. On a tea or rubber estate the reason is easily understood. Children under 10 years earn no money, and consequently it is of small moment if they are ill. Should they contract malaria good parents will arrange for them to get some medicine; indifferent ones will not take the

trouble. Consequently it is not unusual to find in malarious districts anything from 30 to 60 per cent. of the children with parasites in their blood. On the other hand we have proved that they very soon lose their gametocytes, with proper treatment, even without the addition of plasmoquine. With the judicious use of euquinine, mixed with ordinary sweetened condensed milk, it is possible to establish a happy state of affairs among the children on any estate. Euquinine being tasteless, and small children all the world over being greatly attracted to such an excellent sweetmeat as sweetened condensed milk, it becomes easy for a decent dispenser or a native doctor to become so popular that whenever he appears with a bottle all the children in the lines run to him with their mouths open. In a few days, or at the outside in a week, every child on the estate looks out for "the milk." This may be given daily, or two or three times a week, according to circumstances of the case. In the five estates mentioned above, we commenced freeing the children of gametocytes in the middle of February. There was no probability of a large number of anopheles hatching out before the middle of March. The daily dose to every child on the estate rendered them free from gametocytes in their blood before the anopheles appeared. On the estates referred to this treatment was carried on throughout the whole malaria season until the break of the rains. So successful was it that the number of children who sickened with malaria was negligible. It is extraordinary how immune from all manner of disease these children kept under this treatment, even during a moderate outbreak among adults.

As regards anti-gametocyte work for adults, we consider the advent of plasmoquine to be the most important discovery since that made by Sir Ronald Ross of the life cycle of the parasite in the mosquito. We do not propose to quote a large number of authorities in support of the contention that plasmoquine has an action on the gametocytes of malaria. It is so well accepted a fact that it requires no discussion. As soon as this fact was established it was obvious, on mathematical grounds alone, that our position in the fight against malaria was greatly strengthened. We could now wage an anti-gametocyte campaign with some reasonable hope of success, exactly in the same way as the early pioneers commenced anti-larval campaigns. The method we adopted was very simple; we commenced as long ago as 1928. Recognising that plasmoquine was a little-known drug at this time and the necessity for care, we were guided in our choice of methods by the following considerations:—

(a) That the oriental, particularly the Tamil cooly, is much more susceptible to the influence of drugs than the European.

(b) That many of the best coolies on the

estates are girls and young married women whose weight is approximately 6 to 8 stones. It was obvious that small doses were indicated for these.

(c) Further it was very early recognised in Europe and other parts of the world that plasmoquine compound was a comparatively harmless mixture; any accidents that have occurred have been occasioned by the use of plasmoquine pure.

We therefore proceeded on the following lines:—In every patient who sickened with malaria, the treatment was divided into three stages as follows:—

Stage 1.—Whilst the patient is febrile. For this we have given either cinchona febrifuge tablets, or Major Sinton's alkalies and quinine treatment as the circumstances indicated. In the former case this consisted of a morning dose of 2 cinchona febrifuge tablets (each of gr. v.) and an evening dose of one cinchona febrifuge tablet and one tablet of plasmoquine co.; i.e., total for the week 105 grains of cinchona febrifuge and of 0.07 gramme of plasmoquine.

Stage 2.—(Second week). Two tonic pills a day.

Stage 3.—(Third week). Repeat the first week's treatment; total amount of quinine 210 grains, of plasmoquine in 14 doses, 0.14 gramme. This constitutes a three weeks' anti-gametocyte after-treatment. It will be observed that we only give one full dose of plasmoquine co. a day, instead of three as recommended by the makers. For Tamil coolies we find this treatment is invaluable, and we must have given in Ceylon and Southern India at least 100,000 doses and we have never had any complication of a serious nature.

The use of this anti-gametocyte treatment has another profound effect on the influence of health in a labour force. It practically does away with relapses. Everyone conversant with work on estates knows that, in a labour force of say 1,000 in a fairly unhealthy district, there will be something between 50 and 100 "confirmed relapsers," according to the time of the year. These are coolies who suffer from benign tertian malaria and mixed infections; they usually report sick with fever about once a month for 3 to 9 months consecutively. They are a source of considerable loss to the Company owing to inefficiency; they are also a danger to the community, because they are gametocyte carriers. Since the adoption of the anti-gametocyte after-treatment, confirmed relapses are very rare. We could probably do away with them altogether if larger doses of plasmoquine were given, but if this is necessary the patient is sent into the Group Hospital, where he is under observation. For children who are quartan gametocyte carriers, or indeed for any particularly badly infected lot of children, we add small doses of plasmoquine

(pure) to the "euquinine and milk" treatment on two days a week; the addition of plasmoquine does not turn the milk sour or make it any less palatable.

We maintain most emphatically that, with these two new weapons, viz., the anti-gametocyte after-treatment for adults, and the euquinine and milk treatment for children, it is possible to make such an onslaught on gametocyte carriers in a community which will give anything up to 90 to 95 per cent. of success. This fact we have proved by taking blood slides at various times; this will be referred to later on.

Acting on the principles laid down in the foregoing paragraphs, the actual standing orders for the anti-malarial campaign on these estates are as follows:—

Measures against resting anopheles females.

(1) During the winter months of December and January all Europeans', clerks', teamakers', and staffs' bungalows and all servants' quarters near the same are spring-cleaned and white-washed and sometimes fumigated with sulphur. This has prevented the outbreak of malaria, due to hibernating infected females in the neighbourhood of Europeans' and staffs' bungalows which occasionally broke out in the first week in March.

Anti-gametocyte work.

(2) In February all adults are inspected by either the junior writer himself, or the superintendent and the Indian doctor of the estate. All seedy, anæmic looking adults, whether men or women, and particularly new arrivals are put on to a separate list. The malaria register (designed by one of us) gives a list of all cases of malaria that have occurred during the past non-season; the people figuring in the book are specially examined—sometimes blood films are taken; all these unhealthy coolies are given the anti-gametocyte after-treatment as above at least once.

(3) On the 15th February all children on the estate are put on to a daily dose of euquinine and milk. Up to 1929 no exception was made; in 1930 malaria was so rare that only children with spleens and new-comers were treated.

Anti-larval measures.

(4) On the 1st March—or earlier if larvæ are caught in certain warm places on the estate—vigorous anti-larval measures commence; all breeding places are known to the superintendent, and the method of treating them is now thoroughly well known to Kanganies and coolies; oiling is vigorous and thorough; it is carried out weekly throughout the dry season.

(5) If anyone sickens with malaria, his blood film is taken and he is treated as above described.

Incidentally it will be observed that we are great believers in what we may term "the proper timing of measures in a campaign." It

is necessary to study the local conditions of an outbreak carefully and arrange to start measures so that the anopheline population *will never increase beyond the normal winter level*. If the first batch of anopheles that hatch out in the spring is allowed to get into the lines an enormous amount of damage may be done. Our experience on the Tungamullay outbreak confirms this opinion. Extraordinary activity after the event seems to produce very small results. Consequently, therefore, a campaign should be laid down always well ahead of the increase in the anopheline population, and dates must be rigidly adhered to. The writers could never make the Agents of another Company understand the importance of providing the necessary oils, etc., by a certain date; the result was that much of the expenditure was wasted and the results were very poor.

Some time in the early part of 1928 the knowledge of the efficiency of plasmoquine as a drug had advanced sufficiently far to make a test desirable. The subject of the "confirmed relapser" cooly had always been very prominent in our minds, principally because he was such a nuisance, and a definite attempt was made to see whether the new preparation would be of value in doing away with this important source of ill-health and loss.

Before putting our new ideas into practice it was thought desirable to have some definite idea as to the number and kind of parasites that these confirmed relapsers usually harboured. Consequently in 1928 an order was issued by the Manager of the Travancore Tea Company to take blood slides of all coolies on the estates who were known to have relapsed three or four times since the last malarial season which ended with June 1928.

The malarial register kept on these estates rendered this piece of work very easy. As a result of these orders 95 coolies on the estates, who were known to have had relapses were called to the dispensary and the ordinary thick blood films were made from their blood. At the time many of the coolies were not ill. Some were doing an ordinary day's work, but in most cases they were not in a good state of health. On arrival these slides were all stained with Giemsa's stain and examined by one of us. The results are very interesting.

(1) Forty-five per cent. had no parasites at all in their blood.

(2) In about 20 per cent. the corpuscles showed malignant stippling, a shotty appearance, and stained deeply blue.

(3) The results of the remaining 35 per cent. are given in the table. It will be observed that benign tertian largely predominated, as one would expect.

(4) Dividing the parasite carriers into (1) those who had young forms (that is developmental forms, etc.), and (2) those who had gametocytes or both, we find that 41 out of

TABLE II.

*Analysis of parasites in "confirmed relapsers," Travancore Tea Estates Co.'s estates.
August 1928.*

Estates.	Total number of slides.	No plasmodia found and no other evidence of malaria in the blood.	Number without plasmodia but with other evidence of malaria.	Number with parasites.	Young forms only (rings, schizonts, developed forms).	Old forms only (gametes and crescents).	Number with both old and young forms.	Benign tertian.	Malignant tertian.	Quartan.	Mixed (B. T. and M. T.).
Tungamullay ..	18	3	2	13	4	2	7	8	3	..	2
Injikadu ..	13	4	2	7	1	5	1	7	..	..	..
Pasumullay ..	21	5	8	8	4	3	1	7	1	..	..
Mount and Savari ..	6	2	..	4	..	1	3	3	..	..	1
Pambanar ..	14	3	6	5	..	2	3	4	..	..	1
Nellikai ..	8	..	6	2	..	2	..	1	1	..	..
Munjamullay ..	15	2	..	13	2	10	1	7	5	..	1
TOTAL ..	95	19	24	52	11	25	16	37	10	..	5

52 or approximately 80 per cent. of the positive slides showed the presence of gametocytes. This we consider is a very important finding.

The figures demonstrate beyond any doubt the danger to the community which these people give rise to; even though just at that time little harm was done, anopheles were very rare, and the climate cold. Practically all these confirmed relapsers were adults. It should also be borne in mind that a very thorough campaign of anti-malarial measures and treatment of children (though no plasmoquine treatment was then in use) had been in operation during the previous malaria season between March and June of 1928.

During the succeeding months of the autumn

of 1928, viz., September, October, November and December every person on the estates who sickened with fever had his blood taken before his treatment was commenced. At this time the plasmoquine after-treatment was taken into use for the first time, but the important point to remember is that during the previous malarial season or the season of maximum activity March to June of the same year (1928) no plasmoquine treatment was in use. During the autumn 243 slides were sent to us from various estates. These were of course mostly from cases of relapse, reinfections being rare on account of the cold. The results are given below in tabular form, Tables III and IIIa. One or two points are noteworthy.

TABLE III.

*Travancore Tea Estates Company, Limited.
Blood films, Autumn 1928, pre-plasmoquine period.*

Estates.	Total slides.	Negative.	Positive.	B. T.	M. T.	Mix.	Qu.	Gameto-cytes.	Percentage of positive.
Munjamullay ..	83	11	72	28	25	19	..	30	87
Tungamullay ..	28	8	20	16	2	2	..	13	71
Injikadu ..	15	6	9	8	1	..	..	3	60
Nellikai ..	54	14	40	23	12	4	1	13	74
Pambanar ..	4	..	4	4	..	..	..	..	100
Cranby ..	5	5	..	..	..	..	..	..	Nil.
Mount ..	5	2	3	1	1	..	1	1	60
Pasumullay ..	45	17	28	22	3	3	..	23	62
Thengakal ..	4	3	1	..	1	..	..	..	25
	243	66	177	102	45	28	2	83	..

Seventy-three per cent. of all slides positive. Thirty-four per cent. of all slides with gametocytes. Fifty-one per cent. of all positive slides contained gametocytes.

(1) The great prevalence of benign tertian infections.

(2) The high relative frequency of mixed infections. (From long experience of mixed

Altogether there were 235 slides sent from 10 estates. The results are most striking. The cold weather slides numbering 120 only produced 8, or 7 per cent. of positives. Amongst

TABLE IIIa.

Travancore Tea Estates Company, Limited.
Blood films, Autumn 1928, pre-plasmoquine period.

Month.	Total slides.	Negative.	Positive.	B. T.	M. T.	Qu.	Mix.	Gametocytes.	Percentage of positive.
September	45	13	32	17	12	1	2	16	71
October	84	23	61	37	14	..	10	24	73
November	91	23	68	36	15	1	16	34	75
December	23	7	16	12	3	..	1	9	70
	243	66	177	102	44	2	29	83	..

infections, we are of opinion that these cases are particularly difficult to cure. The two varieties of parasites appear to assist each other in keeping alive.)

(3) The percentage of positive slides is about 73, and the percentage of gametocyte carriers in the whole 243 slides is 34 per cent. and 52 per cent. of all positive slides.

In the autumn of 1929 and the spring of 1930 an exactly similar research was carried out on the estates, but during the latter part of 1928 and the whole of 1929 including the fever season, the standard anti-gametocyte after-treatment was administered. Every patient who sickened, and all seedy, anæmic unhealthy looking adults, who were new-comers, received it. The results are given below in Tables IV and IVa.

these there was not a single case which contained matured gametocytes. They were all benign tertian infections. The progressive decline in the percentage of positives month by month should also be noted.

There is, however, an objectionable feature in the figures. No September and October figures are given. Unfortunately the work was only commenced in November. There is no doubt that had September and October figures been available during this year, the percentage of positive slides would have been increased because these slides would be nearer the previous active malarial season. Now compare the two Tables III, IIIa and IV, IVa. Note the drop in the percentage of positive slides from 73 per cent. to 8 per cent. and in the number of gametocytes from 34 per cent. to

TABLE IV.

Travancore Tea Estates Company, Limited.

November 1929 to June 1930, post-plasmoquine period (after-treatment given for a year).

Estates.	Total slides.	Negative.	Positive.	B. T.	M. T.	Mix.	Gametocytes.	Percentage of positive.
Munjamul-lay.	64	45	19	7	11	1	0	30
Tungamul-lay.	114	86	28	14	14	0	4	25
Injikadu	17	10	7	0	7	0	Nil	41
Nellikai ..	15	10	5	0	5	0	1	33
Pambanar	3	3	0	0	0	0	Nil	Nil.
Arnakal ..	6	5	1	1	0	0	Nil	16
Cranby ..	2	2	0	0	0	0	Nil	Nil.
Mount ..	7	7	0	0	0	0	Nil	Nil.
Pasumullay	3	2	1	1	0	0	Nil	33
Thengakal	4	2	2	1	1	0	Nil	50
	235	172	63	24	38	1	..	..

TABLE IVa.

*Travancore Tea Estates Company, Limited.**Blood films, November 1929 to June 1930, post-plasmoquine period (after-treatment given for a year).*

Month.	Total slides.	Negative.	Positive.	B. T.	M. T.	Mix.	Gameto-cytes.	Percentage of positive.
November	25	21	4	4	0	0	0	16
December	11	9	2	2	0	0	0	18
January ..	76	74	2	2	0	0	0	3
February	1	1	0	0	0	0	0	0
March ..	7	7	0	0	0	0	0	0
April ..	23	11	12	4	7	1	3	52
May ..	47	23	24	17	7	0	1	52
June ..	46	26	20	14	6	0	1	44

	Slides.	Positive.	Negative.	Gameto-cytes.	Percentage of positive.
Non-malaria season.	120	8	112	0	7
Malaria season	116	56	60	5	51

nil; this is very striking even making due allowance for the fact that the figures are not quite comparable as regards time of year. There can be no doubt whatever that this improvement in the number of relapses, and in the number of gametocyte carriers is directly due to our anti-gametocyte after-treatment, because all the natural conditions in the two years' experiments are identical. The number of people is about the same, about 9,000 souls; they occupy the same lines on the same estates; in both years (1928 and 1929) anti-larval work took place during the malarial season with equal vigour and with good results in both cases, the only difference being that in 1929 our plasmoquine after-treatment had been in regular use about 12 months, and in 1928 it had not.

We consider that these figures are of great importance and establish beyond all reasonable doubt two things (if it were necessary):—

(1) The efficacy of plasmoquine compound in removing gametocytes from the blood of patients, in preventing relapses; in other words in curing the patients completely.

(2) The great influence that this gametocyte removal—when done systematically—has on the incidence of the disease in any community.

In 1930 malaria practically ceased to exist, so much so that the regular prophylactic dosing of all children was discontinued.

During the malaria season of 1930, three estates, A, B, and C, had the following number of cases during March, April, May, June and July. Vigorous anti-malarial measures were carried out on all, but no plasmoquine co. was administered.

NO AFTER-TREATMENT.

Estates.	Number of cases.	Population.	Number of cases per 100 coolies.
A	239*	820	29
B	84	740	11
C	166	950	18

* 48 serious cases.

Five Travancore Tea Company's estates gave the following results, the latter had used the plasmoquine after-treatment steadily for two seasons.

USED AFTER-TREATMENT.*

Estates.	Number of cases.	Population.	Number of cases per 100 coolies.
Pasumullay	20	650	3
Injikadu	16	475	3
Nellikai	61	700	9
Munjamullay	44	850	5
Tungamullay	38	450	9

* The blood film work shows that only 50 per cent. of these cases were true malaria during the period.

The discussion of the subject of the utility of plasmoquine would not be complete without mention of the work by Barber in the United States of America. He has shown that a ridiculously small dose of plasmoquine prevents the infection of anopheles. The drug appears to completely upset the fertility of the gametocytes and prevents conjugation. The doses we made use of throughout were one-third of what the makers prescribe for malarial patients. The reason for this reduction has been given

fully above. Therefore it follows that the results obtained by us act as a confirmation of Barber's work. Of course, his doses were very small, namely two doses a week. We

give seven doses a week, then an interval of a week, then another seven doses. We understand that Barber's work has been definitely confirmed in India.

TABLE V.

*Travancore Tea Estates Company, Limited.
Anti-malarial campaign, season 1929-30.*

	OIL.			EUQUININE.		CONDENSED MILK.		LABOUR AND SUNDRIES.		TOTAL.		Number of children treated.
	Gals.	Rs.	As. P.	Ozs.	Rs. As. P.	Tins.	Rs. As. P.	Coolies.	Rs. As. P.	Rs. As. P.		
Munjamul-lay.	756	716	10 0	84	157 8 0	192	111 0 0	184	95 8 0	1,080 10 0		161
Injikadu	1,212	1,291	10 5	43	74 6 6	108	63 12 9	375	227 4 0	1,657 1 8		257
Nellikai	768	552	0 0	37	69 14 5	125	72 4 3	126	66 0 6	760 3 2		186
Pasumullay	413	416	0 6	58	108 12 0	103	60 6 9	31½	14 3 10	599 7 1		138
Tungamul-lay.	1,608	1,163	0 7	64	120 0 0	112	63 12 6	458	464 8 11	1,811 6 0		63
	4,757	4,139	5 6	286	530 8 11	640	371 4 3	1,174½	867 9 3	5,908 11 11		805

Abstract.

	Rs.	As.	P.		Rs.	As.	P.
Oil 475 gallons	4,139	5	6	Munjamullay	1,080	10	0
Euquinine 286 ozs. ..	530	8	11	Injikadu	1,657	1	8
Condensed milk 649 tins ..	371	4	3	Nellikai	760	3	2
Sundries	867	9	3	Pasumullay	599	7	1
				Tungamullay	1,811	6	0
TOTAL ..	5,908	11	11	TOTAL ..	5,908	11	11

TABLE VI.

Qualities of crude oil, kerosene oil, euquinine and condensed milk used for each estate and value of same, 1927.

	Crude oil. Gallons.	Kerosene oil. Gallons.	Euquinine. Lbs. Ozs.	Condensed milk. Tins.
Munjamullay	280	200	5 1	277
Tungamullay	70	140	4 9	84
Pasumullay	3	6	1 1	92
Nellikai	120	120	7 6 7	386½
Injikadu	80	160	4 11	165
	553	626	22 12 7	1,004½

	Crude oil.	Kerosene oil.	Transport.	Euquinine.	Milk.	Total.
	Rs. As. P.	Rs. As. P.	Rs. As. P.	Rs. As. P.	Rs. As. P.	Rs. As. P.
Munjamullay	81 10 8	168 12 0	83 7 7	227 13 0	173 2 0	734 13 3
Tungamullay	20 6 8	118 2 0	46 2 11	210 15 0	52 8 0	448 2 7
Pasumullay	0 14 0	5 1 0	2 0 0	46 13 0	57 8 0	112 4 0
Nellikai	35 0 0	101 4 0	45 6 8	333 4 6	241 9 0	756 8 2
Injikadu	23 5 4	135 0 0	52 12 5	210 15 0	103 12 0	525 12 9
	161 4 8	528 3 0	229 13 7	1,029 12 6	628 7 0	2,577 8 9

GRAND TOTAL Rs. 2,577 8 9

Transport calculated at 33.3 per cent.

No report on sanitary measures is really complete without some idea of cost. Therefore we append two tables giving the amount of money expended on the estates which we have treated. The figures in themselves are interesting and may be useful as a basis of comparison with the work of other sanitarians.

Before concluding this note we wish to express our indebtedness to Mr. E. C. Sylvester, the Manager of the Travancore Tea Estates Company in Travancore, for much kind assistance. Any request which we had to make was always met with a ready response and we consider that the success of the measures is to a large extent due to the fact that he has so thoroughly entered into the spirit of the work.

Also to Mr. Cantlay, the Superintendent of Munjamullay, and his wife, we wish to pay a tribute to their keenness and efficiency. The medical arrangements on Munjamullay are the best we have seen in very many years' experience.

Summary and conclusions.

(1) From mathematical reasoning, the only correct way of preventing reinfection in malaria is to reduce the number of anopheles and the number of gametocyte carriers at the same time.

(2) In order to do this, anti-larval measures must commence at the correct moment, the object being to keep the anopheline population down to the normal cold weather level.

(3) Anti-gametocyte operations should precede by approximately a month anti-larval measures.

(4) Euquinine and milk, with or without a little plasmoquine, is the best treatment for removing gametocytes from children.

(5) Suitable doses of plasmoquine compound given as an after-treatment in recovered cases of malaria will greatly reduce the number of gametocyte carriers in any labour force.

If this is not adequate, small doses, two per week, of plasmoquine and quinine given to adults will assist in preventing the disease, by rendering the gametocytes inactive. This fact was originally discovered by Barber in America.

(6) In all highly infected areas anti-larval measures will be necessary for several years. Whether they can eventually be done away with is doubtful, and will depend on the efficiency of the anti-gametocyte treatment.

(7) Anti-gametocyte campaigns are easily carried out and very successful on estates, but present great difficulties among the civil population.

THE TREATMENT OF ORIENTAL SORE WITH BERBERINE ACID SULPHATE.

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IN a previous publication (Das Gupta and Dikshit, 1929) we showed that berberine acid

sulphate inhibits the growth of *Leishmania tropica* even in as high a dilution as 1:80,000, and that it is only slightly irritant to the tissues, causing dilatation of the vessels in the neighbourhood of the injection, followed by simple œdema; it does not cause necrosis of the tissues.

During 1928-29 we treated 6 cases of oriental sore successfully with a 2 per cent. solution of berberine acid sulphate, two to four injections being required to effect a cure.

In order to see whether it would be possible to cure an oriental sore with only one intensive dose of berberine acid sulphate, a suitable patient with 4 lesions on his person was selected.

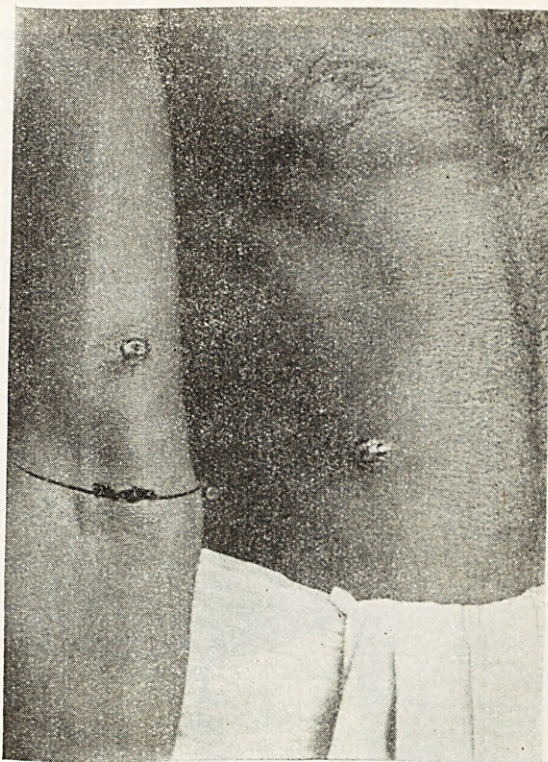


Fig. 1.—Photograph to show lesions 1 and 2 before treatment.

History of the case.—The patient is a healthy young Bengali, who had never been out of Calcutta until two years ago when he went on pilgrimage to Hardwar in the United Provinces, and stayed there 32 days. Six months ago, i.e., 1½ years after his return from Hardwar, he noticed a small papule on the abdomen. When he presented himself for treatment, he showed the following lesions:—

(i) A non-ulcerating nodule, about the size of an 8 anna bit, on the abdomen.

(ii) A similar growth, slightly smaller however, and surrounded by a hyperæmic area about ¼ inch in diameter around the nodule, on the right arm.